

1 sentropy: Similarity-Sensitive Entropy in Python

2 **Phuc Nguyen PhD** ¹, **Rohit Arora PhD** ¹, **Elliot D. Hill MSc** ^{1,2}, **Jasper**
3 **Braun PhD** ¹, **Alexandra Morgan** ¹, **Liza M. Quintana MD** ¹, **Gabrielle**
4 **Mazzoni**^{1,3}, **Ghee Rye Lee MMSc MM** ^{1,4}, **Rima Arnaout MD** ⁵, and
5 **Ramy Arnaout MD DPhil** ^{1,6,7}

6 **1** Department of Pathology, Beth Israel Deaconess Medical Center, Boston, MA 02215, United States **2**
7 Department of Biostatistics and Bioinformatics and Duke AI Health, Duke University School of Medicine,
8 Durham, NC 27710, USA **3** University of Virginia School of Medicine, Charlottesville, VA 22903, USA **4**
9 Department of Surgery, Duke University School of Medicine, Durham, NC 27710, USA **5** Department of
10 Medicine, Bakar Institute for Computational Health Sciences, and Center for Intelligent Imaging,
11 University of California San Francisco, San Francisco, CA 94143, USA **6** Division of Clinical Informatics,
12 Beth Israel Deaconess Medical Center, Boston, MA 02215, USA **7** Harvard Medical School, Boston, MA
13 02115, USA

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Johan Larsson](#) 

Reviewers:

- [@cbueth](#)
- [@richardreeve](#)
- [@thealanjason](#)

Submitted: 26 June 2026

Published: unpublished

License

Authors of papers retain copyright
and release the work under a
Creative Commons Attribution 4.0
International License ([CC BY 4.0](#)).

Summary

Entropy, the main measure of information content, uncertainty, diversity, surprise, and disorder across many fields, derives exclusively from elements' frequencies, ignoring the rich information encoded by elements' similarities. Similarity-sensitive entropy or “sentropy” captures this missing information ([Leinster, 2021](#)). The sentropy Python package calculates traditional entropy's sentropic counterparts, including variants by Rényi order α /Hill viewpoint parameter q (the Rényi entropies, with Shannon entropy for $\alpha = q = 1$), sentropic versions of Hill's diversity indices ([Hill, 1973](#)) (the similarity-sensitive effective/D-number forms of Shannon entropy, Simpson's index, the Berger-Parker index, etc.), and both the Leinster-Cobbold-Reeve (LCR) family of measures and the Vendi scores (quantum sentropies) ([Nguyen et al., 2026](#)). sentropy also calculates sentropic α , β , and γ diversities, relative sentropy (the sentropic generalization of Kullback-Leibler divergence), and cross sentropy. The package has 100% code coverage and is optimized for speed and large datasets.

Statement of need

Shannon entropy and related “traditional” entropies measure the information encoded by the frequency distribution of a system's unique elements. Similarities among elements encode additional information. Some applications:

- **Biological sciences:** immunomes (incorporating sequence, structural, and/or functional similarities between antibodies and between T-cell receptors) ([Arora & Arnaout, 2022](#)), microbiomes (genetic and/or metabolic similarities microorganisms), viromes (sequence and/or tropism similarities among viruses), transcriptomes (RNA transcripts), sequencing libraries (sequences), biomes (species)
- **Medical sciences:** imaging datasets (similarities among images) ([Couch et al., 2024](#)), diseases, patient cohorts (patients), drug candidates, diets (foods)
- **Physical sciences:** materials, chemical compounds, minerals, geography and terrain
- **Engineering/industry:** machine learning (similarities among elements in image/text/video datasets), supply chains, industries (e.g. for antitrust determination), investment portfolios
- **Social sciences:** populations, societies, political organizations, boards of directors, classrooms, committees (to optimize representation), sports teams (similarities among

43 players), museums (exhibits/artwork)

44 The sentropy package is designed to calculate the full range of sentropic quantities in these
45 and other contexts.

46 State of the field

47 Because entropy and diversity are closely related, packages that calculate diversity can be used
48 to facilitate some sentropic measurements. Specifically, packages for calculating frequency-
49 and similarity-sensitive α and β diversities in effective-number form exist for the R (*rdiversity*)
50 and Julia (*Diversity*) programming languages, but not to our knowledge for Python, which is
51 more widely used, especially in machine learning. The Python package *cdiversity* calculates
52 Hill-number diversity but without similarity. The *vendi-score* Python package calculates Vendi
53 scores but not $\alpha/\beta/\gamma$ diversities, relative sentropies, or the LCR framework's quantities.

54 Software design

55 sentropy is designed for speed (via vectorization and parallelization), especially on large
56 datasets. Large datasets present a special challenge: because sentropic measures require
57 calculating the similarity between each pair of unique species in the dataset, and because pairwise
58 similarities are usually inputted as elements of an $n \times n$ matrix (Z), naive implementations
59 tax computer memory for large n , and may even be too large to write to disk. For example,
60 assuming standard 8-byte floating point precision, a single dataset of a million unique elements
61 would require 8TB of memory. Immunomes, microbiomes, and imaging datasets routinely
62 contain tens or hundreds of thousands of unique elements, as do transcriptomes, cell atlases,
63 and other complex datasets. Moreover, a single study can involve tens to hundreds of such
64 datasets (e.g., one per sample/volunteer/patient) and thereby many millions of unique elements
65 in all (for sentropic measurements on the entire collection). *rdiversity* and *Diversity* both require
66 the similarity matrix to be stored in memory. sentropy implements on-the-fly row-by-row
67 calculations for such cases.

68 Research impact statement

69 Since its initial release, sentropy (originally called *greylock*) has been adopted in peer-reviewed
70 scientific literature spanning computational biology and information theory. The package was
71 employed and cited in *Nature Communications* (Ferreira et al., 2025) for self-supervised cardiac
72 ultrasound segmentation, where it supported analysis of over 18,000 clinical echocardiograms.
73 Additionally, it formed the computational backbone of a theoretical investigation into similarity-
74 sensitive entropy metrics published in *Physical Review E* (Nguyen et al., 2026), involving
75 comparative analysis across 53 benchmark ML datasets. sentropy also has applications in
76 data efficiency and data pruning, important in machine learning (Chinn et al., 2023).

77 AI usage disclosure

78 The authors used GitHub Copilot for inline code suggestions. The authors reviewed and
79 validated all AI-suggested edits and bear full responsibility for the final work.

80 References

81 Arora, R., & Arnaout, R. (2022). Repertoire-scale measures of antigen binding. *Proceedings*
82 *of the National Academy of Sciences*, 119(34), e2203505119. [https://doi.org/10.1073/](https://doi.org/10.1073/pnas.2203505119)
83 [pnas.2203505119](https://doi.org/10.1073/pnas.2203505119)

- 84 Chinn, E., Arora, R., Arnaout, R., & Arnaout, R. (2023). ENRICing medical imaging training
85 sets enables more efficient machine learning. *Journal of the American Medical Informatics*
86 *Association: JAMIA*, 30(6), 1079–1090. <https://doi.org/10.1093/jamia/ocad055>
- 87 Couch, J., Arnaout, R., & Arnaout, R. (2024). *Beyond size and class balance: Alpha as a new*
88 *dataset quality metric for deep learning*. <https://arxiv.org/abs/2407.15724>
- 89 Ferreira, D. L., Lau, C., Salaymang, Z., & Arnaout, R. (2025). Self-supervised learning
90 for label-free segmentation in cardiac ultrasound. *Nature Communications*, 16(1), 4070.
91 <https://doi.org/10.1038/s41467-025-59451-5>
- 92 Hill, M. O. (1973). Diversity and evenness: A unifying notation and its consequences. *Ecology*,
93 54(2), 427–432.
- 94 Leinster, T. (2021). *Entropy and diversity: The axiomatic approach*. Cambridge University
95 Press. <https://arxiv.org/abs/2012.02113>
- 96 Nguyen, P., Couch, J., Bansal, R., Morgan, A., Tam, C., Li, M., Arnaout, R., & Arnaout, R.
97 (2026). Comparing frameworks for similarity-aware generalized entropy (sentropy). *Physical*
98 *Review E*, 113(5), 055305. <https://doi.org/10.1103/n2s3-2g85>

DRAFT